N O R M A T I V E

`T = I  det = +1  (x, y) → (x, y)`

"This is the standard."

RFC 2119 MUST / SHALL

Owl Semaphore — Normative

OWL 1 / Identity State / Standard Specification

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SOURCE github.com/IT-Help-San-Diego/owl-semaphore VERSION 1.0 · LICENSE CC-BY-4.0

NORMATIVE — LAYER PROOF PALETTE

1. Statement of Intent

This document defines the **NORMATIVE owl** as a scientific, mathematical, graphical, and archival standard within the Owl Semaphore system.

I am not treating this object as decorative artwork. I am treating it as a **formal epistemic mark** with a defined transform class, a defined coordinate system, a defined layer topology, a defined color and contrast regime, a defined integrity regime, and a defined interpretation rule set.

The goal is simple: the mark must be reproducible, auditable, mathematically coherent, visually stable, and durable across time, tools, and contexts.

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2. System Context

The Owl Semaphore is a visual epistemic notation system based on the Owl of Athena. Its discrete state space is modeled as the Klein four-group:

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\}$$

This is a finite closed subgroup of the orthogonal group $O(2)$.

The NORMATIVE owl is the **identity element** of that system. It is the reference state from which all other states are derived.

This specification concerns only **OWL 1 — NORMATIVE**, but it is written so that the same formal structure can later be extended to OWL 2, OWL 3, and OWL 4.

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3. Ontological Role

3.1 Semantic Designation

NORMATIVE means:

- binding
- canonical
- baseline
- methodological
- reference-grade
- identity-state

3.2 Interpretive Role

This mark is used to signal content that carries the highest internal obligation level in the Owl Semaphore system. In practical terms, that means:

- binding methodology

- canonical specifications
- official operating definitions
- reference interpretations
- baseline measurement standards
- formally adopted system rules

3.3 What It Does Not Mean

The NORMATIVE owl does **not** mean:

- that a claim is metaphysically true
- that a finding is beyond revision
- that a statement is immune from challenge
- that competing interpretations no longer exist

It marks **epistemic status**, not infallibility.

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4. Mathematical Definition

4.1 State Operator

The NORMATIVE owl is defined by the identity transform:

$$T_{\text{norm}} = I$$

4.2 Matrix Form

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

4.3 Determinant

$$\det(I) = +1$$

4.4 Group-Theoretic Class

The NORMATIVE state is:

- orthogonal
- orientation-preserving
- proper
- order 1
- self-inverse only in the trivial sense that $I^{-1} = I$

4.5 Action on Coordinates

For every planar coordinate vector $p = (x, y)^T$,

$$Ip = p$$

This means the NORMATIVE owl is the unique semaphore state that applies no reflection, no inversion, and no rotation to the base reference orientation.

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5. Closure Context

The four-state Owl Semaphore closes under composition because the full discrete set is:

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\}$$

with:

$$\sigma_v \circ \sigma_h = C_2$$

$$\sigma_v \circ C_2 = \sigma_h$$

$$\sigma_h \circ C_2 = \sigma_v$$

The NORMATIVE owl is therefore not merely the first badge in a list. It is the **identity element required for closure, composition, and interpretive reference**.

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6. Coordinate System

6.1 Raster Domain

The canonical raster domain for the current master is:

$$\Omega = \{0, 1, 2, ..., 1079\} \times \{0, 1, 2, ..., 1079\}$$

6.2 Canvas Size

- Width: 1080 px
- Height: 1080 px

6.3 Center

The canonical geometric center is:

$$(c_x, c_y) = (540, 540)$$

6.4 Axis Convention

Raster coordinates are defined as follows:

- x increases to the right
- y increases downward

6.5 Radial Function

For any pixel center (x, y), the radial coordinate is:

$$r(x, y) = \sqrt{(x - c_x)^2 + (y - c_y)^2}$$

All circular and annular zones in the standard are defined relative to this center.

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7. Canonical Orientation

The NORMATIVE owl is the reference pose.

7.1 Orientation Conditions

- upright
- faces right
- no horizontal reflection
- no vertical reflection
- no 180° inversion
- no additional rotational offset

7.2 Transform Interpretation

This is the zero-transform state. Every other semaphore state must be expressible as a transform of this one.

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8. Asset Topology

8.1 Canonical Layer Set

The current NORMATIVE master is defined as a four-layer raster object:

1. **L1 — inner field**
2. **L2 — meander ring**
3. **L3 — owl body**
4. **L4 — outer ring**

8.2 Composite Definition

Let the layers be RGBA-valued fields

$$L_1, L_2, L_3, L_4 : \Omega \to \mathbb{R}^4$$

Then the NORMATIVE composite is defined by ordered alpha compositing:

$$N = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

where $\oplus$ denotes standard ordered alpha compositing on a common canvas.

8.3 Layer Roles

L1 — Inner Field
The base field disk that defines the central tonal ground behind the owl.

L2 — Meander Ring
The shared Greek meander annulus derived from the Athenian tetradrachm geometry.

L3 — Owl Body
The owl figure itself in normative orientation.

L4 — Outer Ring
The outer gold annulus that closes the coin-like boundary and functions as the highest-visibility perimeter accent.

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9. Geometry

9.1 Scaled Geometry Basis

The previously documented 540 px standard defines the following radial structure:

- fill zone: $r \leq 252$
- meander zone: $199 \leq r \leq 256$
- inner hairline: $196 \leq r \leq 202$
- black band: $254 \leq r \leq 256$
- outer ring solid: $258 \leq r \leq 264$
- outer anti-alias edge: $265 \leq r \leq 267$

The 1080 px master is the exact $2\times$ scaling of that system, giving the following nominal radii:

- fill zone: $r \leq 504$
- meander zone: $398 \leq r \leq 512$
- inner hairline: $392 \leq r \leq 404$
- black band: $508 \leq r \leq 512$
- outer ring solid: $516 \leq r \leq 528$
- outer anti-alias edge: $530 \leq r \leq 534$

9.2 Normative Geometry Table

Zone	540 Standard	1080 Master	Function
Fill disk	r ≤ 252	r ≤ 504	Central field
Inner hairline	196–202	392–404	Inner circular accent
Meander annulus	199–256	398–512	Greek key band
Black separator band	254–256	508–512	Ring separator
Outer ring solid	258–264	516–528	Perimeter signal band
Outer AA fringe	265–267	530–534	Anti-aliased edge

9.3 Boundary Logic

The boundary architecture is not incidental. It is part of the standard:

- owl-to-field transition is continuous
- field-to-hairline transition is immediate
- hairline and meander are allowed to overlap by design
- meander and separator are allowed to overlap by design
- separator and outer ring are separated by a narrow transition zone

9.4 Design Principle

The boundary system is intentionally coin-like rather than logo-flat. The structure is annular, nested, and center-locked.

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10. Color Specification

10.1 Normative Palette

The current normative palette is:

- Outer ring: **#d4a853**
- Fill: **#0f0f0f**
- Black band: **#0f0f0f**
- Owl: gold, multi-tonal extracted artwork

10.2 RGB Values

- Gold ring: (212, 168, 83)
- Fill: (15, 15, 15)
- Black band: (15, 15, 15)

10.3 Contrast

Recorded owl-on-fill contrast for NORMATIVE:

$$8.69 : 1$$

This exceeds ordinary minimum UI readability thresholds and is suitable for reference-grade use where visual distinction must be preserved.

10.4 Color Doctrine

For the NORMATIVE owl, gold is not merely decorative. It functions as the visual marker of canonical status, while the near-black field stabilizes contrast and preserves legibility against both white and dark deployment environments.

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11. Transparency and Alpha Regime

11.1 Transparency Rule

The canonical transparent master must satisfy all of the following:

- RGBA mode
- corner alpha = 0
- center alpha = 255
- no false transparency claims in RGB-only files

11.2 Masking Rule

The NORMATIVE transparent composite is defined by a circular alpha mask at the coin edge with anti-aliased feathering.

11.3 Scientific Reason for the Rule

A transparency claim is not accepted as normative unless alpha values verify it. This rule exists to prevent the common design failure mode in which a file is visually placed on a neutral background and then mislabeled as transparent.

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12. Provenance

12.1 Source Statement

The NORMATIVE owl was built from the original NORM composite source and retained the identity transform.

12.2 Transform Statement

- no rotation
- no horizontal flip
- no vertical flip
- no C2 inversion

This is the one state in the system for which the transform chain is intentionally null.

12.3 Transparency Correction

The source file had previously been represented as transparent while still carrying full alpha. The corrected normative master applies an actual circular alpha mask with anti-aliased edge handling.

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13. Asset Invariants

The following are locked invariants for OWL 1 — NORMATIVE.

13.1 Algebraic Invariants

- operator = identity
- determinant = +1
- orientation-preserving
- reference state for all other transforms

13.2 Geometric Invariants

- common center at (540,540) for 1080 masters
- circular annular architecture
- no transform displacement
- no orientation drift

13.3 Visual Invariants

- upright owl
- owl faces right
- gold perimeter band
- dark central field
- shared meander geometry remains intact

13.4 File Invariants

- transparent master must remain RGBA
- composite must be reproducible from canonical layers
- all master assets must remain hash-verifiable

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14. Integrity Regime

14.1 Hash Standard

All normative assets are to be verified using **SHA-3-512**.

14.2 Verification Command

```
openssl dgst -sha3-512 <filename>
```

14.3 Locked Composite Records

The locked 540 px normative composites are recorded as:

- `NORM-composite-white-540.png`
- `NORM-composite-dark-540.png`
- `NORM-composite-transparent-540.png`

with their SHA-3-512 values recorded in the integrity manifest.

14.4 Canonical Source Record

The canonical transparent source-of-truth image for the badge system is recorded as an immutable RGBA PNG in the asset library.

14.5 Integrity Principle

No asset in the normative line is treated as authoritative merely because it looks correct. It must also satisfy:

- geometry compliance
- alpha compliance
- naming compliance
- hash compliance
- provenance compliance

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15. File Naming Standard

15.1 Composite Form

```
NORM-composite-{background}-{size}.png
```

15.2 Layer Form

```
NORM-L{n}-{role}-{size}.png
```

15.3 Master Form

```
NORM-MASTER-{size}.tiff
```

15.4 Permitted Background Tokens

- `white`
- `dark`
- `transparent`

15.5 Principle

Naming is part of the standard. Ambiguous naming is a data integrity problem, not a cosmetic problem.

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16. TIFF Master Policy

16.1 Role of TIFF

The TIFF master is the archival raster container for the normative standard.

16.2 Why TIFF Is Acceptable

TIFF is suitable here because it is:

- durable
- documented
- non-Adobe-dependent
- widely parseable
- appropriate for archival scientific image workflows

16.3 Policy Statement

The TIFF file is the **archival master container**, while the per-layer PNGs remain the most explicit cross-tool representation of atomic layer truth.

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17. Interpretation Rules

17.1 Positive Rule

When the NORMATIVE owl is present, the marked object is to be interpreted as operating under the system's highest ordinary level of internal methodological obligation.

17.2 Negative Rule

The mark does not exempt the underlying content from audit, challenge, correction, or future revision.

17.3 Reference Rule

When other owl states are compared, transformed, audited, or measured, they are compared against the NORMATIVE state as the identity reference.

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18. Non-Permitted Changes

The following changes are non-compliant unless the standard itself is revised:

- rotating the normative owl
- reflecting the normative owl
- changing its center registration
- recoloring the outer ring away from the normative gold assignment
- replacing the inner dark field with an arbitrary fill
- altering the meander geometry without recording a versioned standard change
- flattening away alpha while still calling the file transparent
- renaming assets outside the schema

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19. Relationship to the Other States

The NORMATIVE owl is the base state for all subsequent semaphore transforms.

- NON-NORMATIVE is produced by vertical reflection class σ_v
- CRITICAL is produced by 180° rotation C_2
- METACOGNITIVE is produced by horizontal reflection class σ_h

The normative state therefore has a privileged status: it is not merely one option among four. It is the anchor state required for the interpretability of the rest.

A concrete physical demonstration of the METACOGNITIVE state is the act of searching a room in the ordinary upright position, failing to detect the target, then bending over and viewing the same room upside down through the legs.

20. Physical Instantiation of σ_h: The Through-the-Legs Demonstration

20.1 Operational Description

A concrete physical demonstration of the METACOGNITIVE state is the act of searching a room in the ordinary upright position, failing to detect the target, then bending over and viewing the same room upside down through the legs.

This is not a loose metaphor. It is a direct embodied analogue of the horizontal reflection operator:

$$\sigma_h : (x, y) \mapsto (x, -y)$$

The subject of analysis does not change. The room remains the same room. The search direction remains aimed at the same environment. What changes is the observer's evaluative frame.

20.2 Coordinate Interpretation

Under the ordinary upright search posture, the visual field operates in the baseline orientation. In the simplified local frame used by this system, horizontal structure is preserved while the vertical relation is flipped when the observer bends and looks through the legs:

- left-right structure remains left-right in the operative scan relation
- up-down structure is inverted

This is the defining action of σ_h.

20.3 Why It Works

The practical force of the maneuver is not mystical. It is perceptual.

Human visual processing relies on strong priors:

- light is expected from above
- shadows are expected to fall downward
- gravity-aligned structure is expected to remain gravity-aligned
- habitual scan paths privilege certain object classes, edges, and orientations

These priors are usually useful. They are also filtering mechanisms.

When the observer inverts the vertical axis while keeping the same scene under inspection, the perceptual system's ordinary assumptions no longer fit the incoming image in the usual way. The resulting controlled mismatch exposes features that were being suppressed or deprioritized by fluent upright recognition.

In plain terms: the observer forces the pattern-recognition system to stop coasting.

20.4 Epistemic Meaning

That is why this maneuver is a correct physical analogue for metacognition.

Metacognition is not merely "thinking hard" or narrating one's feelings about thought. It is the deliberate inspection of the machinery of evaluation itself.

In the through-the-legs demonstration:

- the object is unchanged
- the inquiry target is unchanged
- the observer remains the observer
- the evaluative frame is intentionally inverted

This is exactly the logic of the METACOGNITIVE state.

The observer audits the observer.

20.5 Relation to Pattern Recognition

This system is especially relevant in contexts of strong pattern-recognition ability.

High-fidelity pattern recognition is often a strength because it permits rapid discrimination under the baseline frame. But that same efficiency can become a filter. Once the pattern engine has stabilized on a preferred interpretation, it may suppress low-salience anomalies or objects that do not match the dominant recognition template.

The σ_h maneuver interrupts that fluency by feeding the perceptual system vertically inverted data from the same scene. This forces reconstruction rather than automatic recognition.

That reconstruction window is often where the missed object becomes visible.

20.6 Determinant Interpretation

In the semaphore system, identity and 180° rotation have determinant $+1$, while the two reflections have determinant -1.

The through-the-legs maneuver belongs to the reflection class. Its significance is not merely visual inversion but orientation reversal:

$$\det(\sigma_h) = -1$$

This captures the fact that the evaluative frame has been reversed rather than merely preserved under a rotation within the same orientation class.

20.7 Tactical and Applied Implications

The demonstration has implications beyond human introspection.

Any search system that depends on entrenched priors or optimized detection paths can benefit from deliberate frame inversion. This includes:

- human search tasks
- inspection protocols
- interface review
- adversarial analysis
- machine vision and robotics

A robot or vision pipeline that periodically re-examines the same environment under an inversion regime is not being whimsical. It is performing a structured audit of its own detection assumptions.

20.8 Standard Interpretation Rule

When the METACOGNITIVE owl is applied to a document, the intended meaning is:

> the author deliberately inverted the evaluative frame in order to inspect whether the process of perception, judgment, or analysis was itself filtering out something important.

That is the standard meaning. The mark does not say that the subject changed. It says the observer audited the observer.

21. Forward-Looking Rule for the 31° Lean

The measured 31° lean associated with the earlier teal art is not part of the normative state and is not part of the finite closed four-state semaphore set.

That operator is better understood as a processive or transitional gesture rather than a stable epistemic badge state.

For that reason, it shall not be used to define or mutate the normative standard.

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22. Compliance Checklist

An asset claiming to be a compliant **OWL 1 — NORMATIVE** master must satisfy all of the following:

- ☐ identity transform only
- ☐ upright orientation
- ☐ owl faces right
- ☐ no rotation applied
- ☐ no reflection applied
- ☐ common center at (540,540)
- ☐ correct annular geometry
- ☐ gold outer ring (#d4a853)
- ☐ dark field (#0f0f0f)
- ☐ meander geometry unchanged
- ☐ RGBA transparency verified
- ☐ corner alpha = 0
- ☐ center alpha = 255
- ☐ composite reproducible from layers
- ☐ SHA-3-512 hash recorded
- ☐ naming conforms to standard

- ⍰ provenance chain documented

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23. Formal Definition

Let Ω be the 1080×1080 raster domain and let $c = (540, 540)$ be the common image center. Let

$$L_1, L_2, L_3, L_4 : \Omega \to \mathbb{R}^4$$

be the RGBA-valued fields corresponding to inner field, meander ring, owl body, and outer ring, respectively. Define the normative composite image by ordered alpha composition

$$N = L_1 \oplus L_2 \oplus L_3 \oplus L_4$$

with all layers registered to the same center c. Associate to this composite the transform

$$T_N = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

so that for every planar vector p,

$$T_N p = p.$$

Then N is the unique identity-state badge in the semaphore subgroup

$$V_4 = \{I, \sigma_v, C_2, \sigma_h\} \subset O(2),$$

and serves as the reference state for transform, comparison, and interpretive classification.

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24. Closing Statement

This standard treats the NORMATIVE owl as a reproducible scientific-design object rather than a decorative graphic.

That distinction matters.

A decorative logo only has to look right. A normative scientific emblem has to **be** right:

- mathematically
- geometrically
- semantically
- archivally
- and cryptographically

That is the bar.

OWL SEMAPHORE SYSTEM — CLASSIFICATION LEDGER

NORMATIVE

$T = I \quad \det = +1$

$(x, y) \to (x, y)$

"This is the standard."

RFC 2119 MUST / SHALL

NON-NORMATIVE

$T = \sigma_v \quad \det = -1$

$(x, y) \to (-x, y)$

"This reflects the standard."

Informative / Advisory (NOTE)

CRITICAL

$T = C_2 \quad \det = +1$

$(x, y) \to (-x, -y)$

"This inverts the standard."

RFC 2119 MUST NOT / SHALL NOT

METACOGNITIVE

$T = \sigma_h \quad \det = -1$

$(x, y) \to (x, -y)$

"This audits the standard."

Epistemic / Framework (META)

Owl Semaphore v1.0 · github.com/IT-Help-San-Diego/owl-semaphore · DOI: 10.5281/zenodo.19473698
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